

Gokul Bhusal

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Education

Michigan State University

Ph.D., Applied Mathematics

Advisor: Dr. Ekaterina Merkurjev

East Lansing, MI

August 2020 - May 2026

The University of Southern Mississippi

B.S. Mathematics & Minor in Computer Science, Magna cum laude

Advisor: Dr. Zhifu Xie

Hattiesburg, MS

August 2016 - May 2020

Research Interests

Graph-based methods, active learning, optimization algorithms, image analysis (hyperspectral).

Publications

- Delanna Do, **Gokul Bhusal**, Kevin Miller, Stephan Wojtowytsch, Ekaterina Merkurjev, Coverage-based Active Learning with Data-Adaptive Distances (In preparation).
- Delann Do, **Gokul Bhusal**, Kevin Miller, Stephan Wojtowytsch, Ekaterina Merkurjev, VC-Dimension and Generalization in Geometric Machine Learning with Applications to Active Learning (In preparation).
- **Gokul Bhusal**, Yifei Lou, Cristina Garcia-Cardona, Ekaterina Merkurje, A General Framework for Group Sparsity in Hyperspectral Unmixing Using Endmember Bundles, *Journal of Mathematical Imaging and Vision*, 68, no. 3 (2026): 15.
- **Gokul Bhusal**, Kevin Miller, Ekaterina Merkurjev, MALADY: Multiclass Active Learning with Auction Dynamics on Graphs, *IEEE Transactions on Artificial Intelligence* (2025).
- **Gokul Bhusal**, Ekaterina Merkurjev, Guo-Wei Wei, Persistent Laplacian-enhanced Algorithm for Scarcely Labeled Data Classification, *Machine Learning* 113.10 (2024): 7267-7292.
- Zhifu Xie, **Gokul Bhusal**, Hamas Tahir, Central Configurations in the Planar 6-body Problem Forming Two Equilateral Triangles, *Journal of Geometry and Physics*, 153 (2020): 103645. (Undergraduate paper)

Honors and Awards

2025	Douglas A. Spragg Endowed Fellowship - MSU-Math
2024	Outstanding Scholar Fellowship - College of Natural Science, MSU
2024	TA Award for Excellence in Teaching - Department of Mathematics at MSU
2024	Herbert T.Graham Scholarship - MSU-Math
2023	Outstanding Scholar Fellowship - College of Natural Science, MSU
2020	Early Start Fellowship - College of Natural Science, MSU.

Invited Talks/Conference Presentations/Posters

- 2026 SIAM Annual Meeting, Cleveland, Ohio, July 8, 2026 (talk)
- Student Applied Math seminar, Michigan State University, March 18, 2026 (talk)
- Student Applied Math seminar, Michigan State University, April 8, 2026 (talk)
- SIAM Conference on Mathematical and Computational Issues in the Geosciences (GS25), Louisiana, October 14-17, 2025 (talk).
- SIAM Great Lakes Meeting, Illinois Institute of Technology, Sep 27-28, 2025 (poster presentation).

- 14th High Dimensional Data Analysis Workshop (HDDA-XIV), Beaver Island, MI, August 19-22, 2025 (talk).
- SLMATH - IBM Research Summer School, IBM, Yorktown, July 17, 2025 (talk).
- SIAM Conference on Mathematics of Data Science (MDS24), Atlanta, October 21–25, 2024 (poster presentation).
- 2024 SIAM Student Mini-Symposium in Applied Mathematics, University of Michigan, Sep 15, 2024 (talk).
- SIAM Great Lakes Meeting, Michigan State University, Oct 14, 2023 (talk).
- LA/MS Mathematical Associations of America, Loyola University New Orleans, February 2020 (talk).
- USA/USM/SELU Math and Physics Research Mini-Conference, Gulf Park, MS, April 2019 (talk).
- Undergraduate Symposium on Research and Creative Activity, Hattiesburg, MS, March 2018 (poster presentation).
- Joint Mathematics Meeting San Diego, CA, January 2018 (poster presentation).

Summer school and Workshop attended

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| • Joint Mathematics Meeting (JMM), Washington, D.C. | Jan 4-7, 2026 |
| • Workshop: Mathematical Analysis in Machine Learning, University of Pittsburgh | Nov 22-23, 2025 |
| • SLMATH - IBM Research Summer School: Principled Scientific Discovery with Formal Methods (IBM, Yorktown) | July 07-18, 2025 |
| • Workshop: Efficient and Reliable Deep Learning Methods and their Scientific Applications (via Zoom), Banff, Alberta, Canada | June 22-27, 2025 |
| • Workshop: AI, Data and PDE Workshop, MSU | April 1-3, 2025 |
| • Workshop: Fusing Theory and Practice of Graph Algorithms, ICERM | Feb 20-22, 2025 |
| • Optimal Transport Through the Midwest summer school, University of Wisconsin Madison, | July 15-19, 2024. |
| • 88 th Midwest PDE Seminar, The Ohio State University, | April 26-28, 2024 |
| • Winter School in Machine Learning 2024, UT-Austin. | January 15-19, 2024 |
| • Research Experience for Undergraduate (REU) 2019 | June 03 - July 19 |
| • Research Experience for Undergraduate (REU) 2017 | June 19 - August 4 |

Conference/Seminar Organization

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- Co-organizer (with Edem Boahen and Tanvi Mahajan) of Student Applied Mathematics seminar, MSU, Fall 2023 - Spring 2025.

Relevant Skills

Programming languages	MATLAB, C++, Python.
Libraries/Packages	PyTorch, scikit-learn.
Others	HPC Environments, L ^A T _E X.

Services and Professional Organization

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- **Representative of Math department**, Council of Graduate Students (COGS) Fall 2024– Spring 2025
 - **Secretary**, American Math Society, MSU chapter Fall 2023–Spring 2024
 - **Treasurer**, Nepali student Association Summer 2021–Fall 2022

- **Member**, AMS Fall 2020 – Present.
- **Member**, SIAM Fall 2016 – Present.
- **Treasurer** Kappa Mu Epsilon Fall 2018– Spring 2020